Do not write outside the box

0 5 This question is about mixtures.

0 5 . 1 Which substance is a mixture?

[1 mark]

Tick (✓) **one** box.

Air ☐ Gold ☐ Methane ☐ Nitrogen ☐

0 5 . 2 Food colourings are often mixtures of dyes.

What name is given to mixtures that are designed as useful products?

[1 mark]

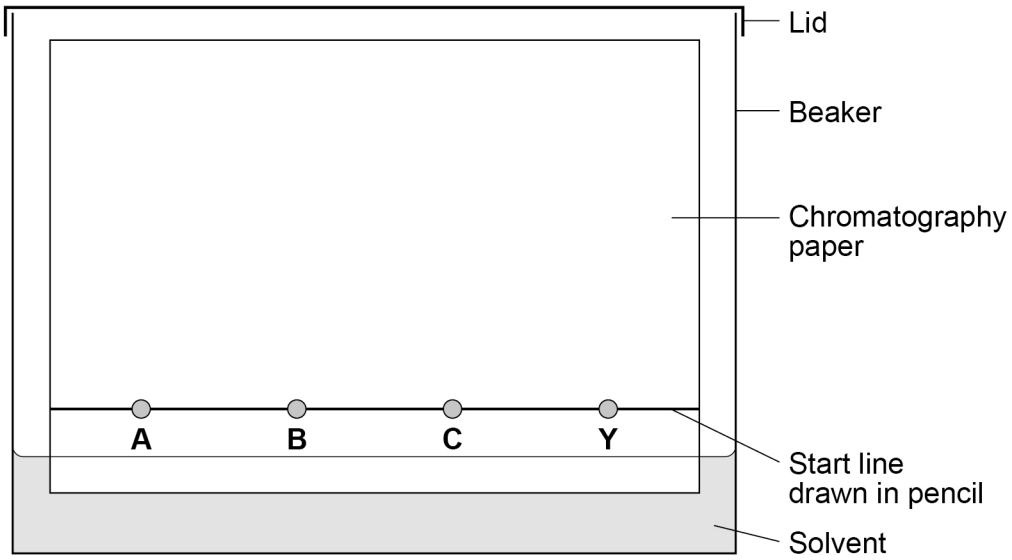
_____

A student investigated a purple food colouring, **Y**, using chromatography.

The student compares **Y** with dyes **A**, **B** and **C**.

0 5 . 3 **Figure 8** shows the apparatus used.

Figure 8



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Chromatography involves a stationary phase and a mobile phase.

Draw **one** line from each phase to what is used for that phase.

Use **Figure 8**.

[2 marks]

Phase	**What is used**
	Beaker
Mobile phase	Chromatography paper
	Food colouring
Stationary phase	Pencil line
	Solvent

Question 5 continues on the next page

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Figure 9 shows the student's results.

Figure 9

0 5 . 4 What **three** conclusions can you make about the dyes in food colouring **Y**?

[3 marks]

1 _____

_____

2 _____

_____

3 _____

_____

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0 5 . 5 In a different experiment a student recorded these results:

Distance moved by dye **G** = 60 mm

Distance moved by solvent = 80 mm

Calculate the R_f value of dye **G**.

$$R_f = \frac{\text{distance moved by dye } \mathbf{G}}{\text{distance moved by solvent}}$$

[2 marks]

R_f = _____

9

Turn over for the next question